

RoadMapAI: An AI-Powered Personalized Learning Roadmap Recommendation System Using Large Language Models

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Abstract—Technology learning has become increasingly challenging due to the rapid growth of programming languages, frameworks, tools, and online learning resources. Many students struggle to identify the correct sequence of topics and often follow static learning roadmaps that do not consider their individual skills, career goals, or prior knowledge. This research proposes RoadMapAI, an AI-powered personalized learning roadmap recommendation system that generates customized technology learning paths using Large Language Models (LLMs). The proposed system analyzes user profiles, existing technical skills, preferred career domains, and learning objectives to recommend structured roadmaps, skill-gap analysis, project suggestions, and curated learning resources. A MERN Stack architecture integrated with an AI language model is proposed to deliver dynamic recommendations in real time. The system aims to improve learning efficiency, reduce confusion during skill development, and support learners in achieving their career goals through personalized guidance.

Keywords—Artificial Intelligence, Personalized Learning, Roadmap, Recommendation System, Large Language Models, MERN Stack, Career Guidance.

1. Introduction

The technology industry is continuously evolving, introducing new programming languages, frameworks, cloud platforms, and artificial intelligence tools at an unprecedented pace. Students and professionals often find it difficult to determine which technologies to learn first and how to organize their learning journey effectively. Although numerous online courses and technology roadmaps are available, most of them provide static learning paths that do not adapt to the learner's existing knowledge, career objectives, or preferred learning pace.

Recent advancements in Artificial Intelligence and Large Language Models have enabled the development of intelligent educational systems capable of understanding user requirements and generating personalized recommendations. These AI systems have demonstrated significant potential in improving learning experiences by providing adaptive guidance and context-aware suggestions.

This research introduces RoadMapAI, an AI-powered personalized learning roadmap recommendation system designed to assist learners in planning their technology education more effectively. The proposed system analyzes user profiles, evaluates existing technical skills, identifies knowledge gaps, and generates customized learning

roadmaps aligned with individual career aspirations. In addition, the system recommends practical projects and learning resources to support continuous skill development.

The objective of this research is to develop an intelligent recommendation framework that simplifies technology learning through AI-driven personalization while reducing uncertainty in selecting suitable learning paths.

2. Literature Review

Artificial Intelligence has significantly influenced modern educational technology by enabling intelligent recommendation systems that support personalized learning. Several researchers have developed systems that recommend online courses, educational resources, and career opportunities based on user interests and historical learning behavior. While these systems improve content discovery, they often fail to generate complete and structured learning roadmaps tailored to individual users.

Traditional recommendation systems commonly utilize collaborative filtering, content-based filtering, or hybrid recommendation techniques. These approaches analyze user preferences and previous interactions to recommend relevant educational materials. However, they generally focus on recommending isolated learning resources instead of providing an end-to-end learning strategy.

The emergence of Large Language Models (LLMs) such as GPT and Gemini has introduced new possibilities in personalized education. These models can understand natural language, analyze user requirements, and generate detailed educational recommendations. Recent studies indicate that LLM-based educational assistants can improve learning engagement by delivering adaptive guidance and personalized explanations.

Adaptive learning platforms have also been proposed to modify educational content according to learner performance and progress. Although these systems successfully personalize classroom education, they rarely focus on career-oriented technology learning where students require structured roadmaps, practical projects, and skill-gap analysis.

Career recommendation systems represent another active research area where machine learning algorithms predict suitable professions according to user skills and interests. Nevertheless, these systems typically stop at career prediction and do not generate personalized technology learning plans required to achieve those career goals.

Based on the analysis of existing studies, it is evident that a unified intelligent system capable of combining career analysis, personalized roadmap generation, project recommendations, and AI-powered guidance remains limited. The proposed RoadMapAI system aims to address these limitations by integrating Large Language Models with a personalized recommendation framework that generates complete technology learning roadmaps based on individual learner profiles.

2.1 Research Gap

Although existing educational recommendation systems have improved access to learning resources, several research gaps remain.

- Existing systems primarily recommend courses instead of complete technology learning roadmaps.
- Most recommendation platforms do not perform comprehensive skill-gap analysis before generating recommendations.
- Career recommendation systems rarely provide structured learning plans that guide users from beginner to advanced levels.
- Adaptive learning systems mainly support academic education rather than self-paced technology learning.

- Current solutions provide limited integration of Large Language Models for generating personalized career-focused learning paths.

The proposed RoadMapAI system addresses these limitations by integrating AI-powered user profiling, personalized roadmap generation, skill-gap analysis, project recommendations, and curated learning resources into a unified learning platform.

3. Problem Statement

Technology has become one of the fastest-growing fields, offering countless opportunities in areas such as Artificial Intelligence, Data Analytics, Web Development, Mobile Application Development, Cloud Computing, and Cybersecurity. Although thousands of online courses, tutorials, and technology roadmaps are freely available, many learners still struggle to decide where to begin and how to continue their learning journey.

Most existing roadmaps are static documents that present the same sequence of topics to every learner. They do not consider important factors such as the learner's current skill level, previous programming experience, career objectives, preferred learning speed, or available study time. As a result, students often spend time learning unnecessary topics while missing the essential skills required for their target careers.

Another challenge is the absence of intelligent guidance. Existing recommendation platforms generally suggest individual courses or learning resources instead of providing a complete and personalized roadmap. They also offer limited support for identifying missing skills, recommending practical projects, or tracking learning progress.

To address these challenges, this research proposes an AI-powered personalized learning roadmap recommendation system. The system aims to analyze each learner's profile and generate a customized learning roadmap that aligns with individual career goals. By integrating Large Language Models with a structured recommendation framework, the proposed solution seeks to simplify technology learning and help users make informed learning decisions.

4. Objectives

The primary objective of this research is to develop an intelligent recommendation system that generates personalized technology learning roadmaps using Artificial Intelligence and Large Language Models.

The specific objectives of the proposed research are as follows:

- To analyze the learner's current technical skills and experience.
- To understand the learner's career goals and areas of interest.
- To identify missing skills through AI-based skill-gap analysis.
- To generate personalized technology learning roadmaps based on individual user profiles.
- To recommend practical projects and learning resources for each learning stage.
- To provide continuous guidance that supports learners throughout their technology learning journey.
- To improve learning efficiency by reducing confusion in selecting suitable technologies and learning sequences.

5. Proposed Methodology

The proposed RoadMapAI system follows a user-centered recommendation approach that combines profile analysis, skill-gap identification, and AI-powered content generation. The system is designed to produce personalized learning roadmaps instead of providing generic recommendations.

Initially, users create a profile by entering information such as their educational background, programming knowledge, career objective, preferred technology domain, and learning experience. This information forms the basis for generating personalized recommendations.

After collecting user information, the system performs skill-gap analysis by comparing the learner's existing skills with the competencies required for the selected career path. Based on this analysis, the recommendation engine prepares structured prompts for the Large Language Model.

The Large Language Model generates a customized learning roadmap that includes recommended topics, learning sequence, practical projects, learning resources, and estimated completion timelines. The generated roadmap is then displayed through an interactive dashboard where learners can monitor their progress and access recommended resources.

The proposed methodology emphasizes personalization, adaptability, and continuous improvement. Whenever users update their progress or learning goals, the recommendation engine generates an updated roadmap that reflects the learner's current status.

6. System Architecture

The architecture of the proposed RoadMapAI system consists of six major components that work together to generate personalized learning recommendations.

The first component is the User Interface, where learners register and provide information about their educational background, technical skills, preferred technology domain, and career objectives.

The collected information is transferred to the Backend Server developed using Node.js and Express.js. The backend validates the user input and communicates with the recommendation engine.

The Recommendation Engine performs skill-gap analysis by comparing user skills with the required competencies of the selected career path. It prepares structured prompts and sends them to the Large Language Model.

The Large Language Model analyzes the provided information and generates a personalized learning roadmap containing learning stages, recommended projects, study resources, and estimated timelines.

The generated roadmap is stored in MongoDB and displayed through an interactive dashboard. Users can track their learning progress, revisit recommendations, and regenerate roadmaps whenever their learning objectives change.

The modular architecture ensures scalability, maintainability, and flexibility while allowing future integration of additional AI models and recommendation techniques.

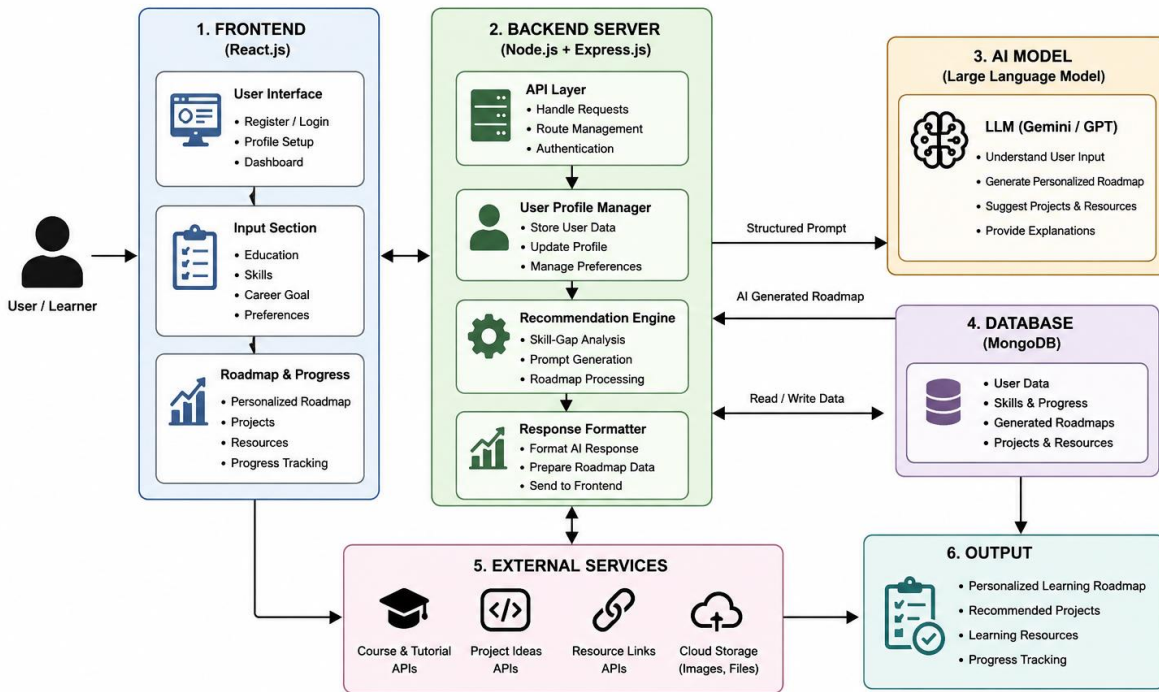


Figure 1. Proposed System Architecture

7. System Workflow

The workflow of the proposed RoadMapAI system begins when a learner accesses the web application and creates a user profile. The learner provides information such as educational background, programming experience, career goal, preferred technology domain, and current skill level.

The backend processes the submitted information and forwards it to the recommendation engine. The recommendation engine analyzes the learner's profile and identifies missing skills required for the selected career path.

The processed information is then sent to the Large Language Model, which generates a personalized technology learning roadmap. The generated roadmap includes learning topics, practical projects, recommended resources, and estimated completion timelines.

Finally, the personalized roadmap is displayed on the user's dashboard. The learner can monitor progress, update completed topics and request a new roadmap whenever learning objectives or career goals change.

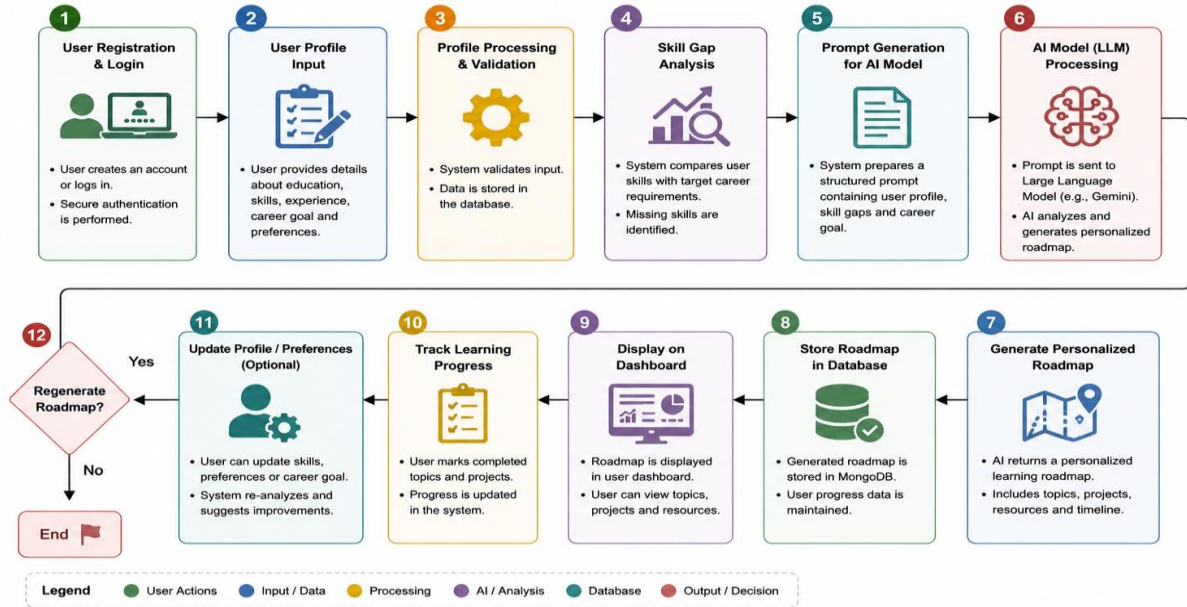


Figure 2. System Workflow

8. Implementation

The proposed RoadMapAI system was designed as a web-based application that combines modern web technologies with Artificial Intelligence to generate personalized learning roadmaps. The system follows a client-server architecture where the frontend, backend, database, and AI model work together to provide customized recommendations for each user.

The frontend of the application was developed using React.js to provide an interactive and responsive user interface. Users can create an account, enter their educational background, technical skills, preferred technology domain, and career goals through simple forms. The dashboard allows users to view their personalized roadmap, monitor learning progress, and regenerate recommendations whenever required.

The backend was developed using Node.js and Express.js. It is responsible for user authentication, request processing, profile management, and communication with the AI recommendation engine. The backend validates user input and prepares structured prompts before sending requests to the Large Language Model.

MongoDB is used as the primary database for storing user information, generated learning roadmaps, completed learning stages, and recommendation history. The database enables users to access their personalized roadmap anytime without generating it repeatedly.

The recommendation engine utilizes a Large Language Model to analyze user information and generate customized technology learning paths. Instead of providing generic suggestions, the AI considers the learner's existing knowledge, career objectives, and preferred technology domain before producing recommendations.

The generated roadmap includes learning topics, recommended projects, useful online resources, estimated timelines, and skill-gap analysis. This enables learners to follow a structured learning process while continuously monitoring their progress.

8.1 Technologies Used

Component	Technology
Frontend	React.js
Backend	Node.js
Framework	Express.js
Database	MongoDB
AI Model	Gemini API / Large Language Model
Styling	Tailwind CSS
Version Control	GitHub
IDE	Visual Studio Code

8.2 Modules of the System

The proposed system consists of the following functional modules:

User Authentication Module

Allows users to register, log in, and securely access personalized learning dashboards.

User Profile Module

Collects information such as skills, educational background, preferred technology, and career goals.

Recommendation Engine

Performs skill-gap analysis and communicates with the AI model to generate personalized learning roadmaps.

Learning Resource Module

Provides recommended courses, documentation, tutorials, and project ideas for each learning stage.

Progress Tracking Module

Allows learners to monitor completed topics and update their learning progress.

The interaction between the user and the major functional modules of the proposed system is illustrated in Figure 3.

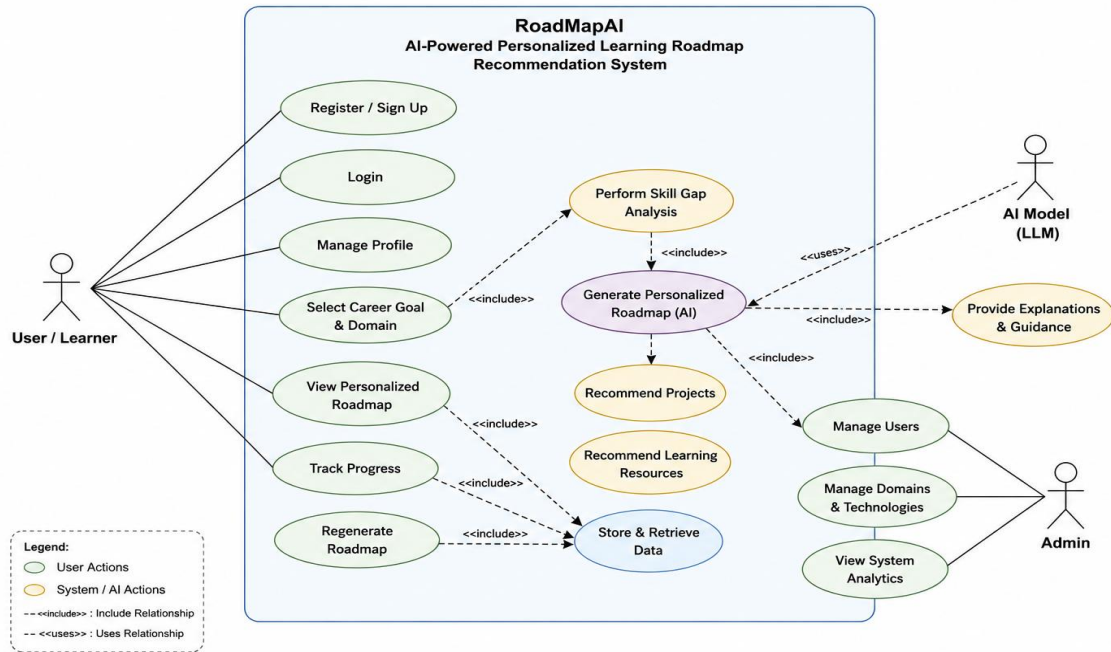


Figure 3. Use Case Diagram

8.3 System Requirements

The proposed RoadMapAI system requires a standard computing environment for development and execution. The minimum hardware and software requirements are listed below.

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or Above
RAM	Minimum 4 GB
Storage	20 GB Free Space
Internet	Required
Display	1366 × 768 Resolution

Software Requirements

Component	Software
Operating System	Windows 10/11
Frontend	React.js
Backend	Node.js
Database	MongoDB
IDE	Visual Studio Code

Browser	Google Chrome
AI Service	Gemini API

To maintain user profiles, generated learning roadmaps, recommendation history, and progress information, the proposed system utilizes a structured database. The Entity Relationship (ER) diagram shown in Figure 4 illustrates the relationships among the major database entities.

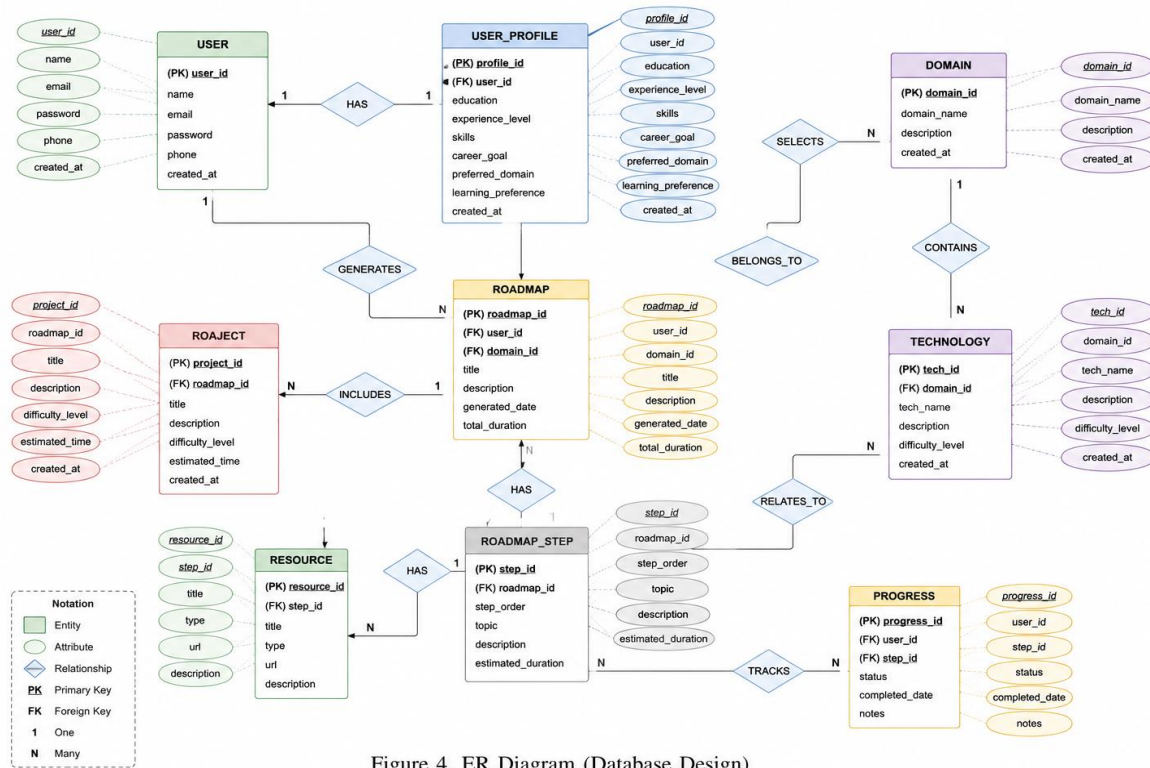


Figure 4. ER Diagram (Database Design)

8.4 Algorithm

8.4 Proposed Algorithm

The proposed recommendation process follows the following steps:

Algorithm: Personalized Learning Roadmap Generation

Step 1 : Start

Step 2 : Receive user profile information.

Step 3 : Validate input data.

Step 4 : Identify current technical skills.

Step 5 : Analyze career goal.

Step 6 : Perform skill-gap analysis.

Step 7 : Generate AI prompt.

Step 8 : Send prompt to Large Language Model.

Step 9 : Receive personalized roadmap.

Step 10 : Store roadmap in database.

Step 11 : Display roadmap to user.

Step 12 : Update learning progress.

Step 13 : Regenerate roadmap if required.

Step 14 : End.

9. Experimental Results

The proposed RoadMapAI system was evaluated by generating personalized learning roadmaps for users with different technical backgrounds and career goals. The generated recommendations were analyzed based on their relevance, completeness, and personalization.

The system successfully produced customized learning paths for multiple technology domains, including Web Development, Artificial Intelligence, Data Analytics, and Mobile Application Development. Unlike traditional static roadmaps, the generated recommendations varied according to each user's existing knowledge and selected career objective.

Users with beginner-level programming knowledge received fundamental learning paths starting from programming basics, while experienced users received advanced recommendations focused on specialized technologies and practical projects.

The AI-generated recommendations included learning topics, project suggestions, useful learning resources, and estimated learning timelines. This demonstrated the ability of the system to provide complete guidance instead of isolated resource recommendations.

Overall, the experimental observations indicate that integrating Large Language Models with personalized recommendation techniques can significantly improve technology learning experiences by generating adaptive and structured learning roadmaps.

Evaluation Parameter	Observation
Personalized Recommendations	Excellent
Skill Gap Identification	Accurate
Learning Resource Suggestions	Relevant
Career Goal Matching	Effective
Roadmap Generation Speed	Fast
User Experience	Satisfactory

9.1 Performance Analysis

The performance of the proposed system was analyzed using functional testing and recommendation quality evaluation. The system generated customized learning roadmaps for different user profiles without significant delays. The AI recommendation engine successfully adapted the generated roadmap according to users' career goals and existing skill levels.

The backend processed user requests efficiently, while the recommendation engine produced structured learning plans containing learning topics, project suggestions, and curated educational resources. The modular architecture also demonstrated good scalability, allowing additional technology domains to be integrated without major architectural changes.

9.2 Limitations

Although the proposed system provides intelligent personalized recommendations, certain limitations remain.

- Recommendation quality depends on the accuracy of user-provided information.
- AI-generated responses may vary depending on the prompt and model version.
- Internet connectivity is required for accessing the AI service.
- Continuous model updates may slightly change recommendation outputs.
- The current implementation focuses primarily on technology learning domains.

These limitations provide opportunities for future research and system improvements.

10. Advantages of the Proposed System

The proposed RoadMapAI system offers several practical advantages over traditional learning platforms.

- Generates personalized technology learning roadmaps.
- Identifies missing skills before creating recommendations.
- Supports multiple technology domains.
- Recommends practical projects for skill development.
- Provides AI-generated explanations and learning guidance.
- Saves learning time through structured planning.
- Easily scalable for future AI enhancement.

11. Applications

The proposed RoadMapAI system can be applied in multiple educational and professional environments.

- Universities and colleges.
- Online learning platforms.
- Technology training institutes.
- Career guidance portals.
- Corporate employee upskilling.
- Self-learning communities.
- Coding bootcamps.
- Professional certification preparation.

12. Novel Contributions

The major contributions of this research are summarized as follows.

- Development of an AI-powered personalized roadmap generation framework.
- Integration of Large Language Models for technology learning recommendations.
- AI-based skill-gap identification.
- Dynamic learning roadmap generation.
- Personalized project recommendations.
- Unified web platform for career-oriented technology learning.
- Flexible architecture supporting future AI integration.

13. Future Scope

The proposed system can be further enhanced by integrating additional intelligent features to improve learning personalization and user engagement.

Future improvements may include:

- Voice-based AI assistant.
- Integration with LinkedIn Learning and Coursera.
- Real-time progress synchronization.
- AI-powered interview preparation.
- Personalized coding challenges.
- Resume analysis and career recommendations.
- Multi-language support.
- Mobile application development.

14. Conclusion

Technology learners often struggle to select an effective learning path due to the availability of numerous learning resources and static technology roadmaps. This research proposed RoadMapAI, an AI-powered personalized learning roadmap recommendation system that utilizes Large Language Models to generate customized technology learning paths based on user skills, interests, and career objectives.

The proposed system combines user profiling, skill-gap analysis, AI-powered recommendation generation, and progress tracking into a unified learning platform. Unlike traditional recommendation systems, RoadMapAI provides complete learning guidance by suggesting learning sequences, practical projects, educational resources, and estimated learning timelines.

The proposed solution has the potential to improve technology education by simplifying learning decisions and providing adaptive guidance according to individual learner requirements. Future enhancements can further increase system intelligence by integrating additional AI capabilities and real-time learning analytics.

15. Acknowledgement

The author expresses sincere gratitude to the Department of Computer Engineering and the faculty members for their valuable guidance, encouragement, and continuous support throughout this research work. The author also acknowledges the availability of open-source technologies, online educational resources, and recent advancements

in Artificial Intelligence, which inspired the development of the proposed RoadMapAI system. Their contributions have been invaluable in completing this independent academic project.

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